جامعة عفت
EFFAT UNIVERSITY

## Smart Sustainable Mobile Laundry

# BarqWash

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**OSCM 306**

# Overview of BarqWash









# About BarqWash

What is BarqWash?

- Mobile on-site laundry service in Saudi Arabia
- Washing, drying, ironing, folding at customer location
- Fast, convenient, sustainable service
- Supports Vision 2030



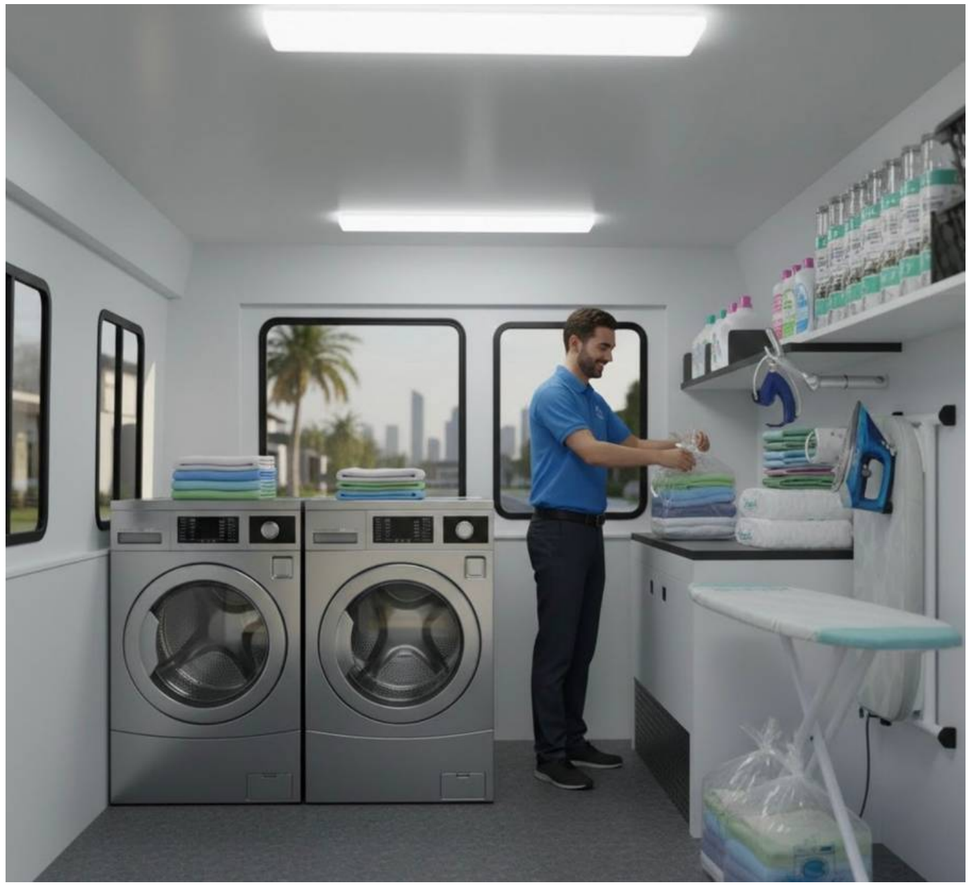
# Procurement Objective

**Goal of This Project**

- Select the best suppliers for key operational products:
- Cargo Van
- Washing Machine
- Dryer
- Eco-Friendly Detergent
- Packaging Materials

**Method Used:**

- Weighted supplier evaluation model.



# Procurement Evaluation Methodology

This structured methodology ensured objective and data-driven procurement decisions.



1. **Identify Products**
   Cargo van, washer, dryer, detergent, packaging
2. **Select Suppliers**
   Local, regional, and global options
3. **Collect Data**
   Pricing, quotations, delivery, service
4. **Apply Weighted Scoring**
   Criteria-based supplier evaluation
5. **Recommend Suppliers**
   Best supplier selected per product

# Evaluation Criteria

| Criteria | Weight |
|---|---|
| Cost | 20% |
| Quality | 15% |
| Delivery | 15% |
| Reliability | 10% |
| Customer | 10% |
| Sustainability | 10% |
| Innovation | 5% |
| Risk | 5% |
| Technical | 10% |



# Product 1: Cargo Van

| Supplier | Vehicle | Result |
|---|---|---|
| Toyota Hiace | Reliable | Good |
| Ford Transit | Spacious + customizable | Best |
| Maxus G10 | Low cost | Basic |



Ford Transit

**Recommended Supplier:** Mohamed Yousuf Naghi Motors (Ford Transit)

**Reason:**

Best operational flexibility despite higher cost.

# Supplier Scoring Model

## BarqWash Vans

| | | LOCAL | REGIONAL | GLOBAL | | | |
|---|---|---|---|---|---|---|---|
| **Criteria** | **Weight (%)** | **(Naghi) Score (1-100)** | **(Petromin) Score (1-100)** | **(Ford) Score (1-100)** | **Weighted Score (A)** | **Weighted Score (B)** | **Weighted Score (C)** |
| **Cost and Pricing** | 15.00% | 70 | 85 | 70 | 10.5 | 12.75 | 10.5 |
| **Product Quality** | 20.00% | 95 | 85 | 95 | 19 | 17 | 19 |
| **Delivery and Logistics** | 15.00% | 90 | 85 | 75 | 13.5 | 12.75 | 11.25 |
| **Customer Service** | 10.00% | 90 | 90 | 85 | 9 | 9 | 8.5 |
| **Supplier Reliability** | 15.00% | 90 | 90 | 90 | 13.5 | 13.5 | 13.5 |
| **Innovation and Customization** | 15.00% | 98 | 88 | 95 | 14.7 | 13.2 | 14.25 |
| **Risk Management** | 10.00% | 88 | 85 | 90 | 8.8 | 8.5 | 9 |
| **Total** | 100% | | | | 89 | 86.7 | 86 |

| | |
|---|---|
| **Supplier A** | **89** |
| **Supplier B** | **86.7** |
| **Supplier C** | **86** |

# Product 2: Washing Machine

**Suppliers compared:**

- Al-Halabi
- LG Electronics
- Alibaba

| Supplier | Score | Rank |
|---|---|---|
| Al-Halabi | 82.7 | 1 |
| LG | 82.4 | 2 |
| Alibaba | 55.1 | 3 |

**Winner: Al-Halabi Laundry Equipment**

**Why:**

- Fast delivery
- Installation included
- Low risk
- Strong local support



# Supplier Scoring Model

## Washing Machines

| | | LOCAL | REGIONAL | GLOBAL | | | |
|---|---|---|---|---|---|---|---|
| Criteria | Weight (%) | Al-Halabi Laundry Equipment Score (1-100) | LG Electronics Score (1-100) | Alibaba Supplier 2 Score (1-100) | Weighted Score (A) | Weighted Score (B) | Weighted Score (C) |
| Cost and Pricing | 10.00% | 72 | 80 | 85 | 7.2 | 8 | 8.5 |
| Product Quality | 10.00% | 85 | 94 | 60 | 8.5 | 9.4 | 6 |
| Delivery and Logistics | 10.00% | 95 | 78 | 45 | 9.5 | 7.8 | 4.5 |
| Customer Service | 10.00% | 92 | 80 | 50 | 9.2 | 8 | 5 |
| Supplier Reliability | 10.00% | 93 | 84 | 55 | 9.3 | 8.4 | 5.5 |
| Sustainability Practices | 10.00% | 65 | 82 | 50 | 6.5 | 8.2 | 5 |
| Innovation and Customization | 10.00% | 68 | 85 | 80 | 6.8 | 8.5 | 8 |
| Risk Management | 10.00% | 90 | 75 | 40 | 9 | 7.5 | 4 |
| Technical Support and Service | 10.00% | 95 | 78 | 42 | 9.5 | 7.8 | 4.2 |
| Innovation and Continuous Improvement | 10.00% | 72 | 88 | 65 | 7.2 | 8.8 | 6.5 |
| Total | 100% | | | | 82.7 | 82.4 | 57.2 |

| | |
|---|---|
| Supplier A | 82.7 |
| Supplier B | 82.4 |
| Supplier C | 57.2 |

# Product 3: Commercial Dryer

**Compared:**

- Local
- Regional
- International suppliers

**Best Supplier: Cleantech Gulf**

**Why:**

- Best balance of cost + quality + service
- Highest score (81.05)



# Supplier Scoring Model

## Commercial Dryer

| | | LOCAL | REGIONAL | GLOBAL | | | |
|---|---|---|---|---|---|---|---|
| Criteria | Weight (%) | CHIBAB Score (1-100) | Cleantech Gulf Score (1-100) | Alliance Score (1-100) | Weighted Score (A) | Weighted Score (B) | Weighted Score ( |
| Cost and Pricing | 20.00% | 75 | 85 | 70 | 15 | 17 | 14 |
| Product Quality | 20.00% | 70 | 78 | 90 | 14 | 15.6 | 18 |
| Delivery and Logistics | 15.00% | 95 | 82 | 65 | 14.25 | 12.3 | 9.75 |
| Customer Service | 10.00% | 90 | 85 | 75 | 9 | 8.5 | 7.5 |
| Supplier Reliability | 10.00% | 85 | 80 | 90 | 8.5 | 8 | 9 |
| Sustainability Practices | 10.00% | 65 | 75 | 85 | 6.5 | 7.5 | 8.5 |
| Innovation and Customization | 5.00% | 60 | 80 | 85 | 3 | 4 | 4.25 |
| Risk Management | 5.00% | 90 | 80 | 70 | 4.5 | 4 | 3.5 |
| Technical Support and Service | 3.00% | 90 | 85 | 85 | 2.7 | 2.55 | 2.55 |
| Innovation and Continuous Improvement | 2.00% | 60 | 80 | 88 | 1.2 | 1.6 | 1.76 |
| Total | 100% | | | | 78.65 | 81.05 | 78.81 |

| | |
|---|---|
| Supplier A | 78.65 |
| Supplier B | 81.05 |
| Supplier C | 78.81 |

# Product 4: Eco-Friendly Laundry Detergent

**Compared:**

- Saudi suppliers
- Regional suppliers
- Global suppliers

**Recommended: Al Emlaq Eco Clean**

**Why:**

- Local availability
- Low supply chain risk
- Eco-friendly formula
- Supports sustainability goals






# Supplier Scoring Model

## Laundry Detergrent

| | | LOCAL | REGIONAL | GLOBAL | | | |
|---|---|---|---|---|---|---|---|
| Criteria | Weight (%) | Al Emlaq (Local) | Wiz Trading (Reg | Seventh Generation | Weighted Score (Lo | Weighted Score (R | Weighted Score (Global) |
| Cost and Pricing | 20% | 85 | 92 | 88 | 17.00 | 18.40 | 17.60 |
| Product Quality | 15% | 82 | 80 | 95 | 12.30 | 12.00 | 14.25 |
| Delivery and Logistics | 15% | 95 | 78 | 70 | 14.25 | 11.70 | 10.50 |
| Customer Service | 10% | 90 | 82 | 88 | 9.00 | 8.20 | 8.80 |
| Supplier Reliability | 10% | 90 | 80 | 90 | 9.00 | 8.00 | 9.00 |
| Sustainability Practices | 10% | 85 | 75 | 98 | 8.50 | 7.50 | 9.80 |
| Innovation and Customiz | 5% | 70 | 78 | 92 | 3.50 | 3.90 | 4.60 |
| Risk Management | 5% | 95 | 70 | 65 | 4.75 | 3.50 | 3.25 |
| Technical Support and Se | 5% | 75 | 75 | 90 | 3.75 | 3.75 | 4.50 |
| Innovation and Continuo | 5% | 70 | 72 | 92 | 3.50 | 3.60 | 4.60 |
| Total | 100% | | | | 85.55 | 80.55 | 86.90 |

| | |
|---|---|
| **Supplier A** | **85.55** |
| **Supplier B** | **80.55** |
| **Supplier C** | **86.9** |

# Product 5: Packaging Materials (Laundry Bags)

**Suppliers Compared:**

- Napco (Local)
- Hotpack (Regional)
- AliExpress (Global)

**Recommended Supplier:**

**Hotpack**

**Why Selected:**

- Highest total score (87.70)
- Strong product quality
- Reliable delivery performance
- Good customization options

| Supplier | Score | Rank |
|---|---|---|
| Hotpack | 87.7 | 1 |
| Napco | 86.75 | 2 |
| AliExpress | 78.05 | 3 |





# Supplier Scoring Model

**plastic garment covers (laundry bags)**

| | | LOCAL | REGIONAL | GLOBAL | | | |
|---|---|---|---|---|---|---|---|
| **Criteria** | **Weight (%)** | **Napco Score (1-100)** | **Hotpack Score (1-100)** | **Aliexpress Score (1-100)** | **Weighted Score (A)** | **Weighted Score (B)** | **Weighted Score (C)** |
| **Cost and Pricing** | 20.00% | 85 | 85 | 85 | 17 | 17 | 17 |
| **Product Quality** | 15.00% | 90 | 88 | 80 | 13.5 | 13.2 | 12 |
| **Delivery and Logistics** | 15.00% | 90 | 90 | 80 | 13.5 | 13.5 | 12 |
| **Customer Service** | 10.00% | 85 | 90 | 80 | 8.5 | 9 | 8 |
| **Supplier Reliability** | 10.00% | 90 | 88 | 78 | 9 | 8.8 | 7.8 |
| **Sustainability Practices** | 10.00% | 85 | 85 | 65 | 8.5 | 8.5 | 6.5 |
| **Innovation and Customization** | 5.00% | 85 | 88 | 75 | 4.25 | 4.4 | 3.75 |
| **Risk Management** | 5.00% | 90 | 90 | 70 | 4.5 | 4.5 | 3.5 |
| **Technical Support and Service** | 5.00% | 80 | 88 | 75 | 4 | 4.4 | 3.75 |
| **Innovation and Continuous Improvement** | 5.00% | 80 | 88 | 75 | 4 | 4.4 | 3.75 |
| **Total** | 100% | | | | 86.75 | 87.7 | 78.05 |

| | |
|---|---|
| **Supplier A** | **80.25** |
| **Supplier B** | **87.7** |
| **Supplier C** | **78.05** |

# Procurement Risks

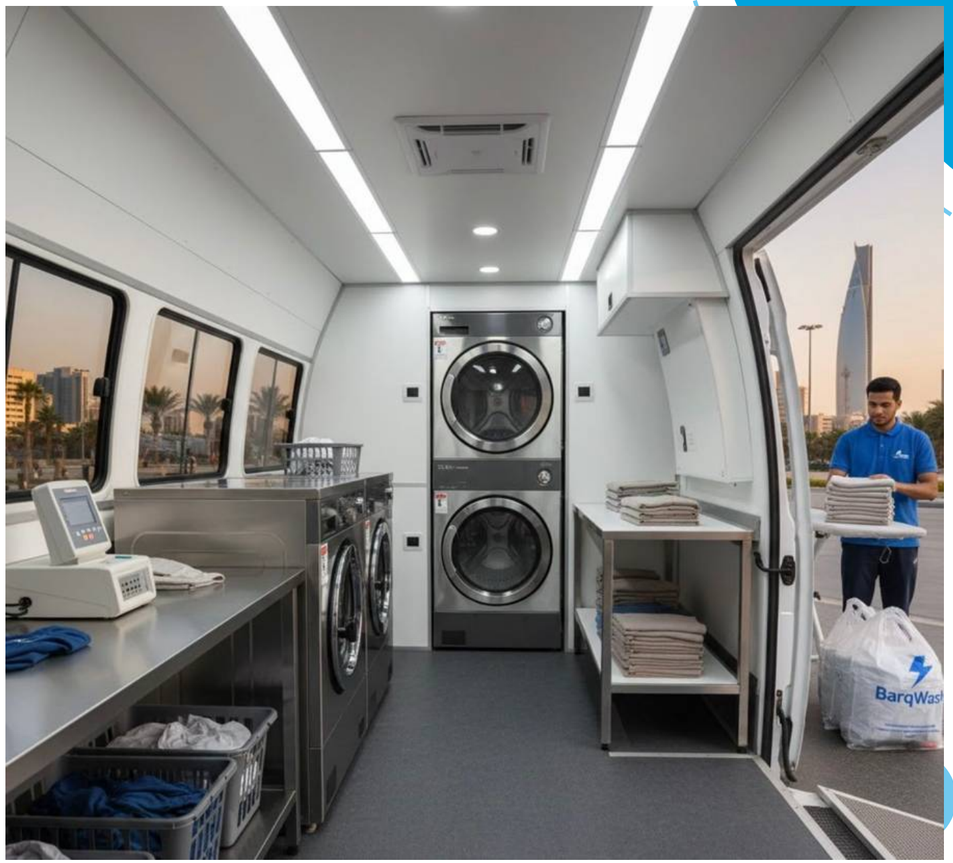
**Main Risks:**

- Delivery delays
- High maintenance cost
- Fuel price changes
- Customs/import delays
- Product quality inconsistency

**Mitigation:**

- Dual sourcing
- Buffer stock
- Local backup suppliers
- Long-term contracts

| Risk | Level |
| --- | --- |
| Delivery | High |
| Fuel | Medium |
| Quality | High |



# Key Findings

**Local suppliers strongest in speed & support** → **International strongest in innovation** → **Mixed sourcing model recommended**

**Regional suppliers balanced value**

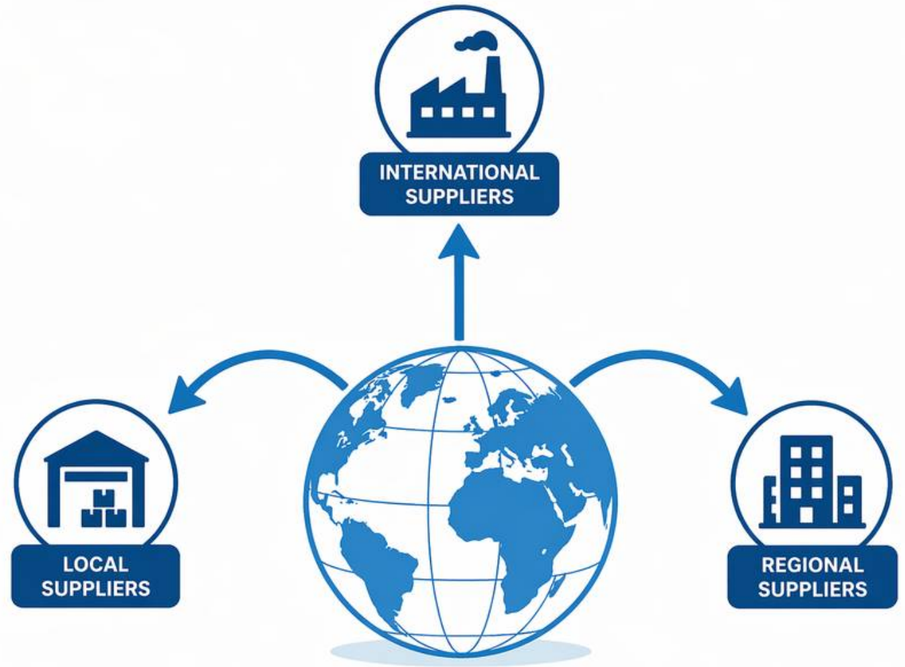




A diversified supplier strategy offers the strongest balance of cost, speed, innovation, and reliability.

# Final Supplier Strategy

| Product | Recommended Supplier |
|---|---|
| Cargo Van | Naghi Motors |
| Washer | Al-Halabi |
| Dryer | Cleantech Gulf |
| Detergent | Al Emlaq |




- Strategy Summary
- Local sourcing for critical daily items
- Regional sourcing for equipment value
- International sourcing selectively
- Reduce risk through diversification

# Conclusion

The procurement analysis successfully identified the best suppliers for BarqWash’s key operational needs.

A weighted supplier evaluation model helped ensure objective and data-driven decisions.

The final sourcing strategy balances cost, quality, reliability, and sustainability.

Using a mix of local, regional, and international suppliers reduces risk and improves operational performance.

This strategy supports BarqWash’s long-term growth and service excellence.

# THANK YOU